

House Price Prediction – Project Summary

This project aimed to predict house prices using machine learning techniques based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities. The Housing dataset was explored, cleaned, and preprocessed by handling duplicates and converting categorical variables into numerical form using one-hot encoding.

Two regression models were trained and evaluated: Linear Regression and Random Forest Regressor. The dataset was split into 80% training data and 20% testing data. Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

The Linear Regression model achieved an MAE of approximately 970,043, an RMSE of 1,324,507, and an R^2 Score of 0.653, outperforming the Random Forest model on this dataset. This indicates that the Linear Regression model was able to explain about 65% of the variation in house prices.

Correlation analysis showed that area, bathrooms, air conditioning, stories, and parking were the most influential features affecting house prices. Larger houses with more bathrooms and modern amenities generally had higher market values. Additionally, unfurnished houses tended to have lower prices compared to furnished properties.

Based on these findings, real estate businesses should focus on highlighting property size, parking availability, air conditioning, and other key amenities when marketing homes, as these factors significantly influence property value.